



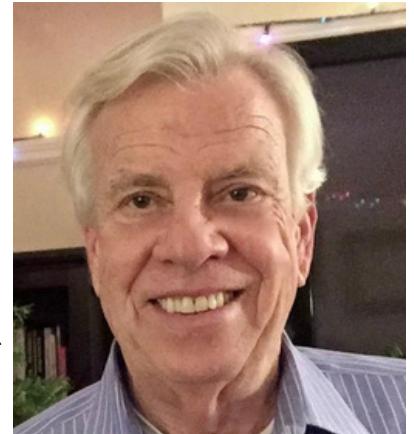
PAStimes

A Publication of the Phoenix Astronomical Society

Stu's Views • Stu Brackney, President PAS

Hi everyone. December and January were busy months for PAS. Lectures; star parties and of course telescope workshops. A few reminders:

Telescope Loaner Program for Members Only: Do you own a telescope? Do you want to try using a telescope? If so, contact telescope@pasaz.org because for two weeks you can borrow one of our many scopes that have been donated to us. This is a brand-new program being offered to PAS members only.



PAS Membership is only \$30.00. If you like what we are doing, tell a friend. Bring a friend to our lectures, star parties and of course telescope workshops. That includes members of your family.

Are you selling your telescope? The board over a year ago discontinued advertisements in our PAStimes. Some of you have asked that it be returned. Your board is reviewing that topic. Expect to hear soon about your board's decision.

Link of the month: <https://astronomy.tools/> This is one amazing source of information for astronomers, from star charts, cloud forecasts to "field of view calculators" and viable astronomy links.

Your Board of Directors and Managers are keeping remarkably busy, but we still need one more volunteer to step up to fill the position of Event Manager. For details, contact me: stuartbrackney@pasaz.org. Clear skies everyone. Stu

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February Upcoming Events

February 1st – Speaker Meet & Greet Dinner

February 1st – Monthly Lecture Series

February 10th – Public Star Party

February 15th – Telescope Workshop 101 – What to Buy

February 15th – Telescope Workshop 201 – Operating Your Telescope

For more information and to register for these events, go to the [PAS Event Calendar and Registration Site](#). It is important to register for all events. In case of an event change or cancellation, you will be notified via email.

February Lecture Series

The speaker for our meeting on Thursday, February 1, 2024, is Dr. Gregory Brenneka. The title of the presentation is: “Meteorites: Their incredible influence on our planet, culture, and origin.”



Meteorites were the literal building blocks of the planets; they later provided the raw materials to kick-start life on Earth and meteorites have had an inordinate amount of influence on life once it got started. Meteorites also allow scientists to look back in time to figure out how the Solar System formed and evolved. He will also talk about how meteoritic bodies are investigated to reveal the original structure of the Solar System.

Prior to joining the Lawrence Livermore National Laboratory in 2019 as a Staff Scientist, Dr. Brenneka led the Solar System Forensics group at the University of Münster (Germany) for five years. He has a PhD in geochemistry from Arizona State University, an MS from Oregon State University, and BS degrees from the University of Missouri and Columbia College. There will not be a “Meet and Greet the Speaker” dinner before the meeting because he will be presenting remotely from California.

For more details and to register for the meeting and presentation go to the PAS Event Calendar and Registration Site:

[PAS Event Calendar and Registration Site](#)

It is important to register for events. In case of changes, you will be notified.

January 4, 2024, Presentation by Mather Ota by Paul Facuna

January 4, 2024 presentation had to be cancelled because the presenter could not be contacted to confirm he would be presenting. Instead, Stuart Brackney, president of the Phoenix Astronomical Society, gave an overview presentation on PAS. Some points he covered were:

- **Monthly speaker presentations**
- **Star Tours Group – what it does and how to become a member**
- **Monthly telescope workshops for members and non-members**
- **Where and what is on the event calendar**
- **Necessity of registering for events**
- **The new Telescope Loan Program for members**
- **Looking for a volunteer to be Event Coordinator**

January Outreach Events

The PAS outreach program offered several excellent events in January.:

Public Star Party

A good-sized group of people visited our Apache Wash Trailhead star party site on Saturday the 13th for our first public star party of the new year. 11 PAS members attended with their telescopes and provided a wide variety of excellent views of our cosmos to their fellow members and the visiting public. The evening was a little cloudy and provided the first hint of the cooler temperatures ahead but was enjoyed by all.

Star Tours – Heros Regional Park Library

The Star Tours outreach program held an event for the Heros Regional Park Library in Glendale on the 8th. Richard Sauerbrun served as PAS's EAA telescope operator for the event which included over 60 guests. Sky conditions were excellent, and Richard operated two telescopes with two display screens to entertain the actively engaged crowd with views of several celestial targets.



Telescope Workshop 101 – What to Buy

January's 101 workshop on the 18th at PVCC was taught by Stu Brackney. There were two attendees who were looking for assistance with buying a telescope.

Telescope Workshop 201 – How to Operate My Telescope

January's 201 workshop on the 18th at the Apache Wash Trailhead was taught by Damon Finnel, Richard Pipkin, and Mason Klenklen. A total of five students showed up with their telescopes to improve their observing skills.

Star Tours – ASU NASA SciAct STEM Meeting

The Star Tours outreach program supported a NASA-sponsored event on the ASU campus in Tempe. Mark Johnston, Andrew Stone, and Scott Adams served as telescope operators for the event.

Star Tours – Desert Harbor Steam Night

The Star Tours outreach program held an event for Desert Harbor Elementary School in Peoria. Roger Anzini, Damon Finell, Chris Ford, and Mason Klenklen served as PAS's telescope operators for the event.



Star Tours Outreach Program's roster of presenters for 2024

Mark Johnston
Ted Blank
Mason Klenklen
Mark Savan
Andrew Stone-Star Tour Manager
Richard Sauerbrun
Stu Brackney
Bruce Wurst
Roger Anzini
Peter Turner
Chris Ford
Tony Pistilli
Scott Adams
Damon Finell

These individuals have successfully met the qualifications to participate in Star Tours Events and receive compensation for their services. 🌟

Member Astro Images



angel nebula
(ngc 2170)



Mark Savan

heart nebula
(ic 1805)



Mark Savan



rosette
nebula



Richard Sauerbrun



SUN



Mark Johnston

jupiter



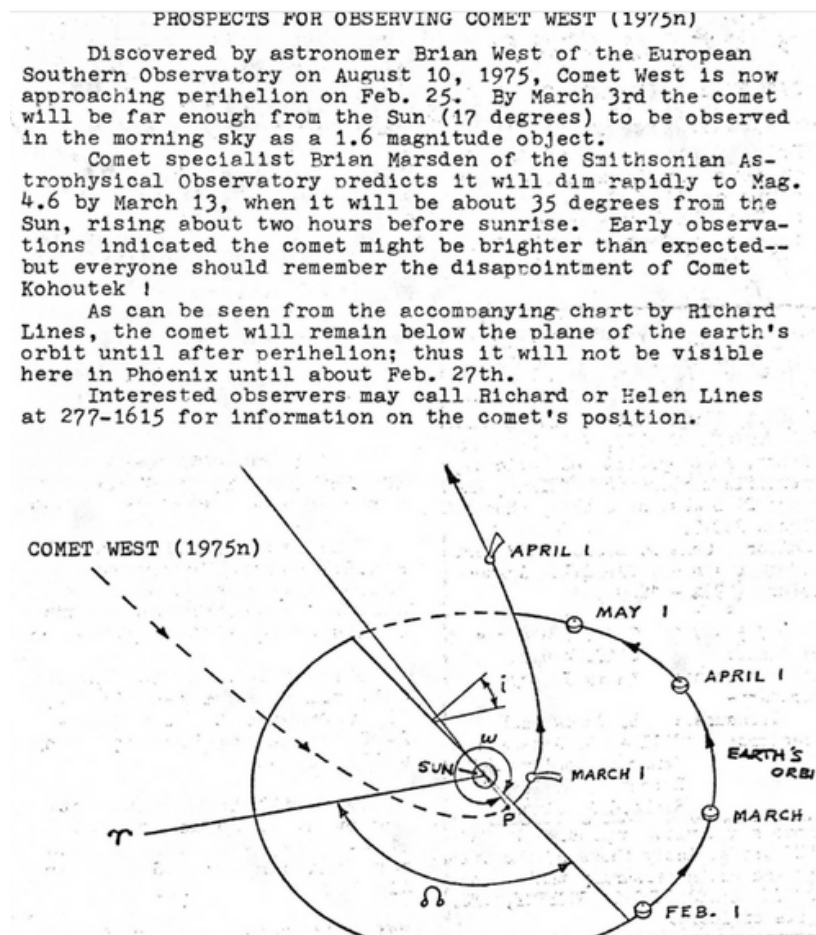
Mark Johnston



Small Telescope Targets for February

Each month we feature 7 good targets for small telescopes that should be visible in the evening.

1. M45 The Pleiades Famous asterism in Taurus
2. M41 star cluster Open cluster in Canis Major about the size of the full Moon
3. Canis Major 145 The Winter Albireo
4. Castor triple A 6-star system, three of which are visible in small scopes
5. Galactic Anticenter This is not something to see, but just beside Beta Auriga is the opposite direction from the center of the Milky Way
6. M36, 37, 38 Trio of nice open star clusters in Auriga
7. Beta Monoceros Beautiful triple star in Monoceros



Blast from the past- a snippet from the
February 1976 newsletter.

Helpful Astronomy Websites

by Mark Johnston

The following websites are great free resources to help you plan and enjoy your astronomy adventures

Astromart astromart.com/ This is one of the top two sites for buying and selling used astronomy gear, product reviews, and forums where you can discuss any aspect of astronomy with other enthusiasts. A small fee is required.

Astrometry.net <https://nova.astrometry.net/> This site will take any astronomy image you send it (galaxy, starfield, nebula, etc) and then tell you what it is and where it is in the sky.

Astronomy Tools astronomy.tools/calculators/field_of_view/ This site has a lot of capabilities but I use it primarily to figure out how a target (e.g. the Helix nebula) will look in my eyepiece or astronomy camera. You pick your scope, camera, eyepiece and target and the site does the rest.

Astrospheric www.astrospheric.com/ Probably the best short term astronomy weather forecasting tool, providing predictions of clouds, transparency and seeing.

Clear Sky Charts www.cleardarksky.com/ Excellent source of astronomy seeing conditions forecast. Also a great tool for finding observing sites with lower light pollution.

Cloudynights www.cloudynights.com/ This is one of the top two sites for buying and selling used astronomy gear, product reviews, and forums where you can discuss any aspect of astronomy with other enthusiasts. Free.

Dark site finder darksitefinder.com/map/ Great website for finding a dark observing location.

Heavens above heavens-above.com/ Top site for predicting naked eye observations of satellites. See when the ISS is visible in your neighborhood!

Jupiter transits shallowsky.com/jupiter/ For anyone with a small telescope, this is a great site. You can find out when the Great Red Spot and Jupiter's satellite transits are visible.

Solar Dynamic Observatory (SDO) <https://sdo.gsfc.nasa.gov/data/> Find out how the Sun looks right now, at various wavelengths. If you own a solar telescope, this site coupled with Astrospheric is all you need to plan fabulous solar observing!

Telescopius telescopius.com/ This site is very helpful in planning an observing evening. It will tell you when any particular target is rising and setting and when is the best time to observe it. This allows you to sequence your observing in a planned manner.

Additional Information

PAS Sam Insana Student Scholarship winner at Paradise Valley Community College will soon be announced. Your membership fees allow us to do this each year.

Members Only Telescope Loaner Program

GREAT news everyone, we are launching our new Members Only Telescope Loaner Program.

The entire PAS inventory of telescopes is available for active members to use. Some telescopes require the completion of a training program, others do not.

Several of the loaner telescopes represent a significant dollar value. In certain cases, the PAS member may be asked to provide credit card information as insurance against potential damage or loss.

All loaner telescopes include a kit of eyepieces and accessories. There is also a special kit of advanced 2" eyepieces available for loan. Loan periods are for a maximum of two weeks. However, longer periods may be granted in special cases.

All telescope equipment and accessories must be picked up at the PAS storage facility and delivered back to the facility by the end of the loaner period.

If you wish to check out one of PAS's telescopes, send an email to telescope@pasaz.org with the following information:

- Full Name
- Email Address
- Desired Telescope and/or accessory kit
- Check out date
- Check-in date
- Additional information as needed

